



Information Technology Institute (ITI)

CMOS Analog IC Design

Lab 11: Fully-Differential Folded Cascode OTA

(Mini Project 2)

Lab Report

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Part 1: g_m/I_D Design Charts

NMOS Design Charts

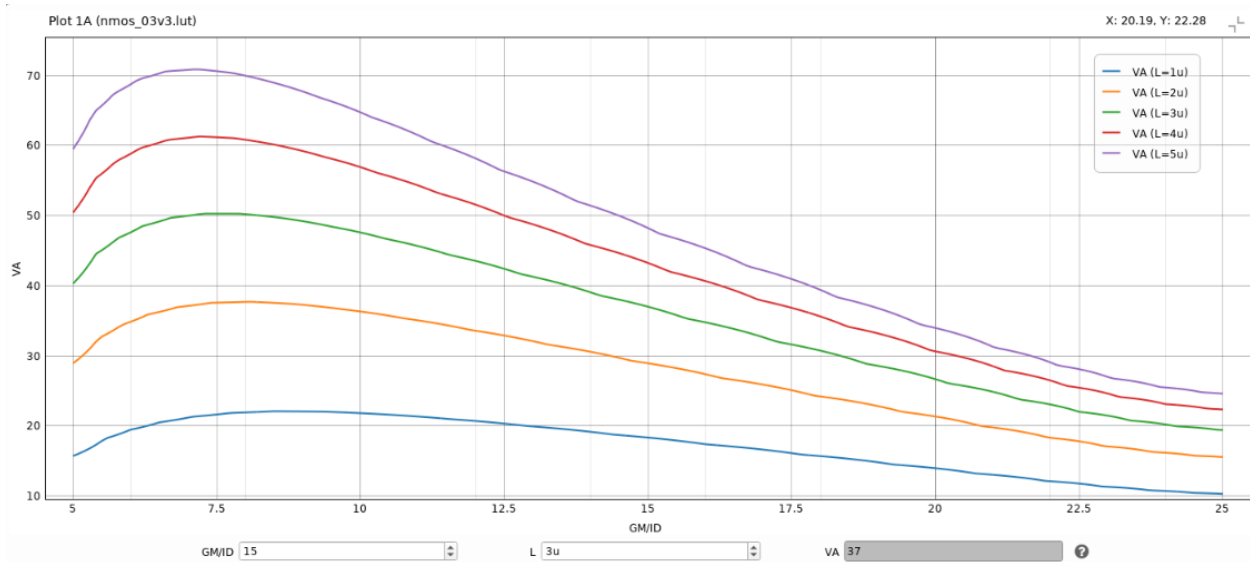


Figure 1 NMOS: V_A vs g_m/I_D

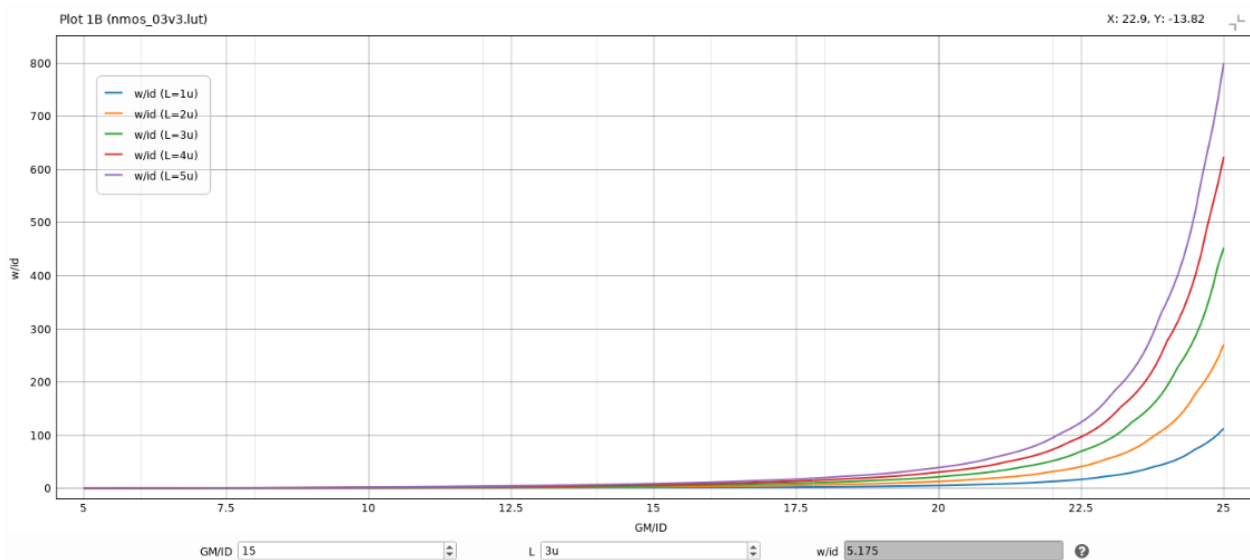
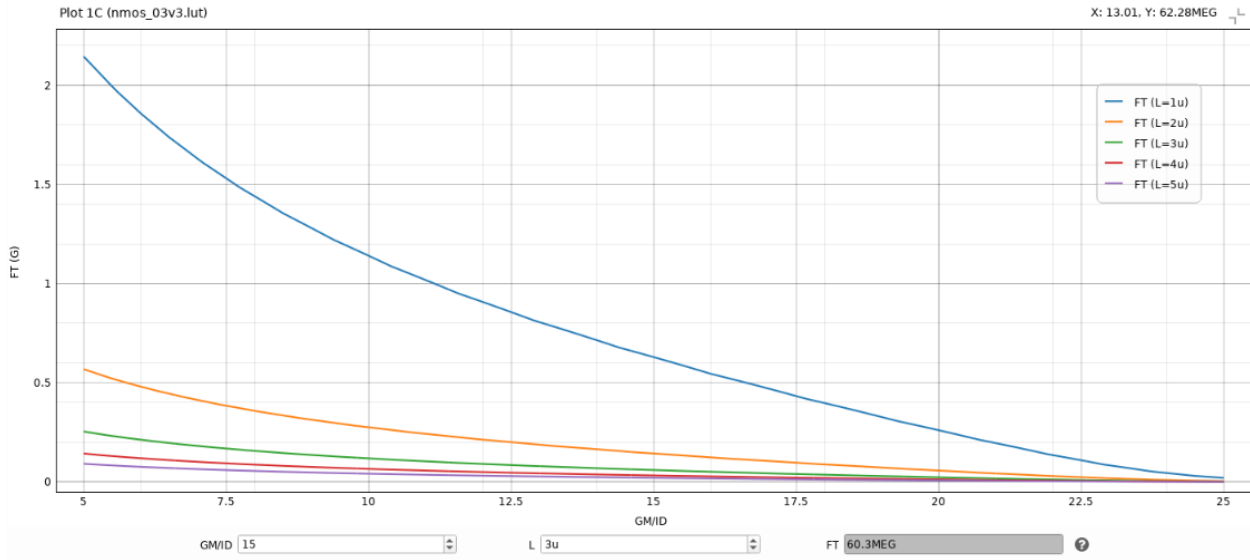
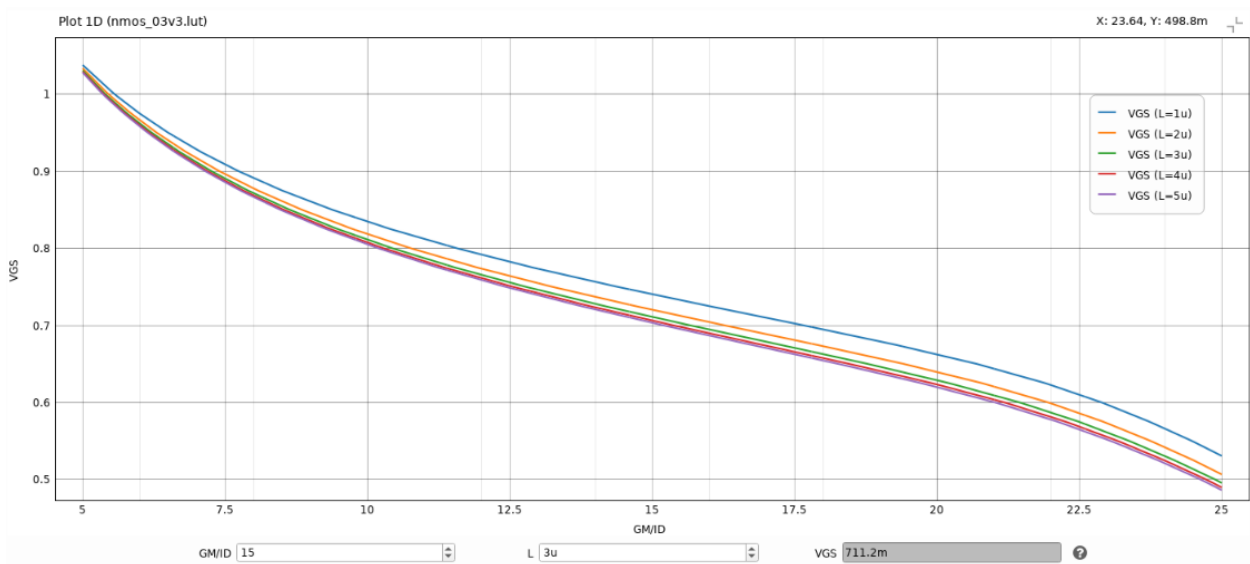


Figure 2 NMOS: W/I_D vs g_m/I_D

Figure 3 NMOS: f_T vs g_m/I_D Figure 4 NMOS: V_{GS} vs g_m/I_D

PMOS Design Charts

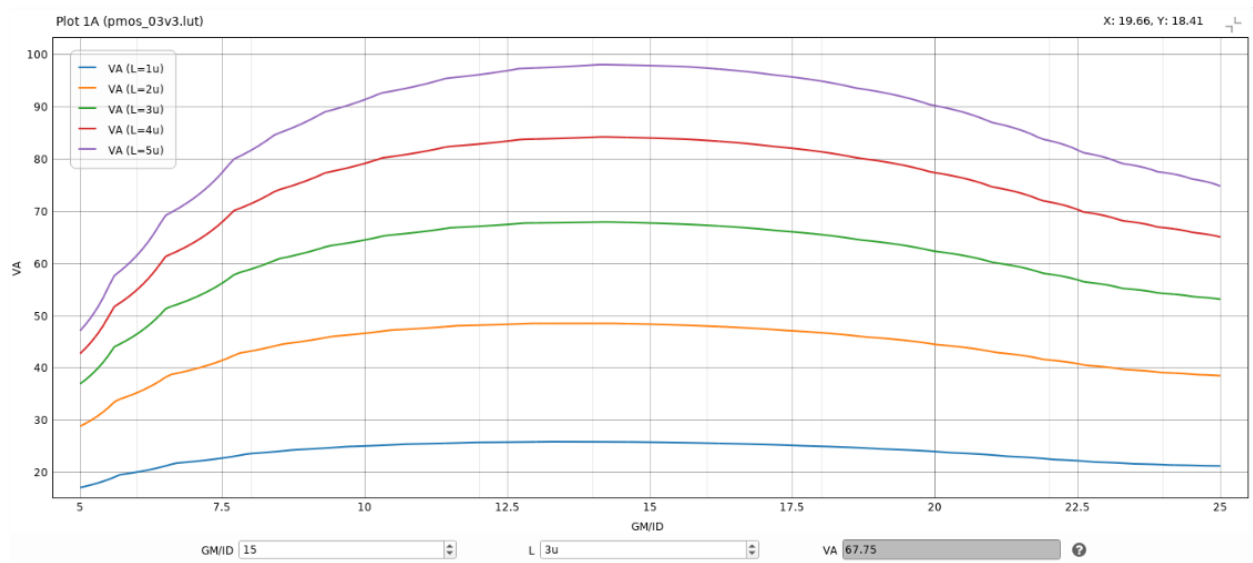


Figure 5 PMOS: V_A vs g_m/I_D

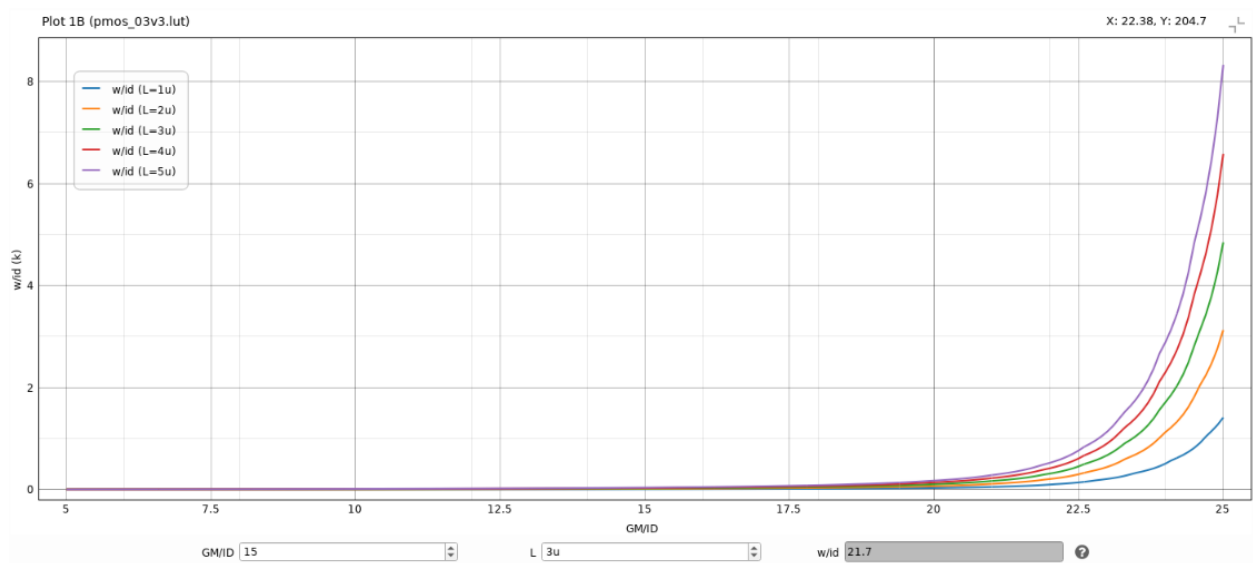
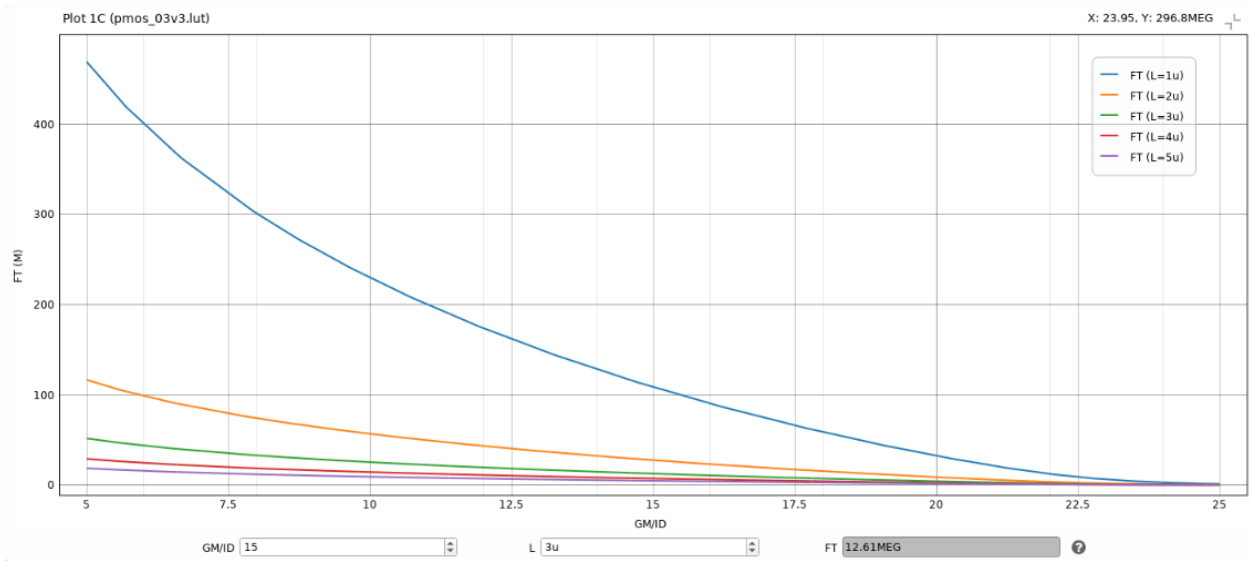
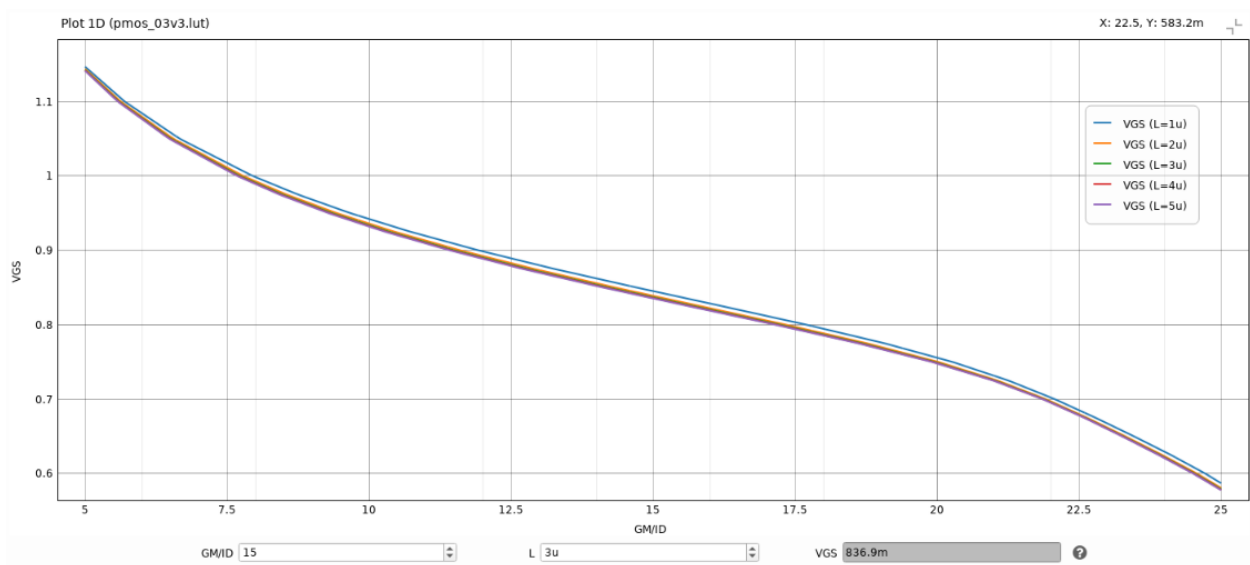


Figure 6 PMOS: W/I_D vs g_m/I_D

Figure 7 PMOS: f_T vs g_m/I_D Figure 8 PMOS: V_{GS} vs g_m/I_D

Part 2: OTA Design

Design Specifications

Parameter	Value	Unit
Supply Voltage (V_{DD})	2.5	V
Closed Loop Gain	2	V/V
Phase Margin	≥ 70	$^{\circ}$
CMIR Low	≤ 0	V
CMIR High	≥ 1	V
Differential Output Swing	1.2	V _{pk-to-pk}
Load Capacitance (C_L)	500	fF
DC Loop Gain	≥ 60	dB
Settling Time (1% error)	≤ 100	ns

Table 1: Folded cascode OTA design specifications

Architecture Selection

A fully-differential folded cascode OTA with capacitive feedback is designed. The architecture consists of:

- **Input stage:** PMOS differential pair (folded)
- **Cascode stage:** NMOS and PMOS cascode devices
- **Load:** PMOS current source loads
- **CMFB:** Behavioral CMFB (Part 3), Actual CMFB (Part 4)

Input Pair Selection

From the CMIR spec (≤ 0 V and ≥ 1 V), a **PMOS input pair** is used since it can operate at low common-mode voltages.

Folded Cascode OTA circuit

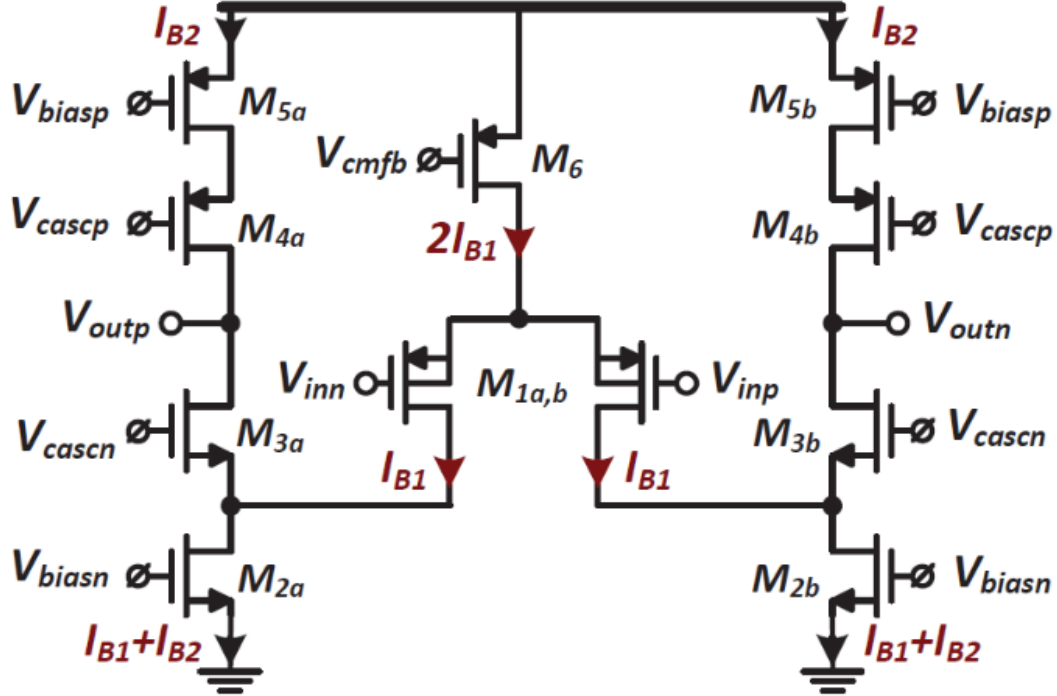


Figure 9 Folded cascode OTA

Design Procedure

Settling Time to Bandwidth Conversion

For a first-order system with 1% settling time:

$$t_s = \frac{\ln(100)}{2\pi BW_{CL}} = \frac{4.605}{2\pi BW_{CL}}, \quad BW_{CL} \geq \frac{4.605}{2\pi \times 100 \times 10^{-9}} = 7.33 \text{ MHz}, \quad BW_{CL} \geq 7.33 \text{ MHz}$$

gm input pair Calculation

$$GBW_{CL,min} = \frac{g_{m1}}{2\pi C_{out,total}} = \frac{1}{2\pi \tau \beta} = 21.96 \text{ MHz}$$

$$C_{out,total} = \frac{C_F \times C_{in}}{C_F + C_{in}} + C_L = \frac{1 \times 2}{1 + 2} + 0.5 = 1.167 \text{ pF}$$

$$g_{m,in} \geq 2\pi \times GBW \times C_{out,total}$$

$$g_{m,in} \geq 161 \mu S$$

Assumed Transistor Parameters

Based on folded cascode trade-offs:

Transistor Type	L (μm)	g_m/I_D (S/A)	Region
Input Pair (PMOS)	0.44	16.5	MI/WI
Cascode (NMOS)	0.3	8	SI
Cascode (PMOS)	0.28	8	SI
Current Sources (PMOS)	1	8	SI
Current Sources (NMOS)	1	8	SI

Table 2: Assumed transistor parameters

With four devices stacked between V_{DD} and ground in the folded cascode the assumption $V_{DS} = V_{DD}/4 = 0.625$ V for all transistors

PMOS & NMOS Current Mirrors

For current mirrors small g_m/i_d needed to get lower matching and Area so i assumed $g_m/i_d=8$

▼ LUT Settings

LUT: nmos_03v3 ?

Corner: TT ☐ All ?

Temp (°C): 27.0 ☐ All ?

Frequency: 1 ?

ID: 10u ?

g_m/I_D : 8 ?

L: 1u ?

VDS: 0.625 ?

VSB: 0 ?

Stack: 1 ?

Results:

Name	TT-27.0
3 L	1u
4 W	2.76u
5 VDB	625m
6 VGB	892.3m
7 VGS	892.3m
8 VDS	625m
9 VSB	0

▼ LUT Settings

LUT: pmos_03v3 ?

Corner: TT ☐ All ?

Temp (°C): 27.0 ☐ All ?

Frequency: 1 ?

ID: 10u ?

g_m/I_D : 8 ?

L: 1u ?

VDS: 0.625 ?

VSB: 0 ?

Stack: 1 ?

Results:

Name	TT-27.0
3 L	1u
4 W	11.84u
5 VDB	625m
6 VGB	998.9m
7 VGS	998.9m
8 VDS	625m
9 VSB	0

(a) NMOS Current mirror

(b) PMOS Current mirror

Figure 10 NMOS and PMOS cascode transistor sizing using SA

Input Pair (M1A/M1B)

The input pair is biased at $g_m/I_D = 16.5$ to avoid the exponential area increase that occurs beyond $g_m/I_D = 17$ while $r_o = 1\text{ M}\Omega$ is chosen to be comparable to the current mirror $r_o = 1.2\text{ M}\Omega$ to prevent gain reduction.

▼ LUT Settings

LUT ?

Corner ☐ All ?

Temp (°C) ☐ All ?

Frequency ?

ID ?

gm/ID ?

ro ?

VDS ?

VSB ?

Stack ?

Results:

	Name	TT-27.0
3	L	440n
4	W	33.48u
5	VDB	625m
6	VGB	820.2m
7	VGS	820.2m
8	VDS	625m
9	VSB	0

Figure 11 NMOS Cascode (M3/M4)

PMOS & NMOS Cascode Transistors

For Cascode transistors $gm/id = 8$ for minimum area selection Lmin was used for PMOS cascode device while $L=300nm$ for NMOS cascode to get comparable r_o results

LUT Settings		
LUT	nmos_03v3	?
Corner	TT	All ?
Temp (°C)	27.0	All ?
Frequency	1	?
ID	5u	?
gm/ID	8	?
L	300n	?
VDS	0.625	?
VSB	0.625	?
Stack	1	?
Results:		
Name	TT-27.0	
3 L	300n	
4 W	480n	
5 VDB	1.25	
6 VGB	1.591	
7 VGS	966m	
8 VDS	625m	
9 VSB	625m	

LUT Settings		
LUT	pmos_03v3	?
Corner	TT	All ?
Temp (°C)	27.0	All ?
Frequency	1	?
ID	5u	?
gm/ID	8	?
L	min	?
VDS	0.625	?
VSB	0.625	?
Stack	1	?
Results:		
Name	TT-27.0	
3 L	280n	
4 W	1.36u	
5 VDB	1.25	
6 VGB	1.732	
7 VGS	1.107	
8 VDS	625m	
9 VSB	625m	

(a) NMOS Cascode (M3/M4) (b) PMOS Cascode (M5/M6)

Figure 12 NMOS and PMOS cascode transistor sizing using SA

$$VCASCN \approx V_{GSN} + V_{DS} = 0.966 + 0.5 = 1.466 \text{ V}$$

$$VCASCP \approx V_{DD} - |V_{GSP}| - V_{DS} = 2.5 - 1.107 - 0.625 = 0.768 \text{ V}$$

$$r_{o,pc} = 774.2 \text{ k}\Omega$$

$$r_{o,nc} = 686 \text{ k}\Omega$$

Sizing Table

Table 3: Folded cascode OTA transistor parameters

Parameter	M1A/M1B	M2A/M2B	M3A/M3B	M4A/M4B	M5A/M5B	M6	M7	M8	M9
Type	PMOS	NMOS	NMOS	PMOS	PMOS	PMOS	NMOS	PMOS	PMOS
Role	IP	NMOS CM	Casc N	Casc P	PMOS CM	PMOS CM	NMOS CM	PMOS CM	I_{ref}
I_D (μ A)	10	15	5	5	5	20	10	5	10
g_m/I_D (S/A)	16.5	8	8	8	8	8	8	8	8
V^* (mV)	121.2	250	250	250	250	250	250	250	250
L (μ m)	0.44	1	0.3	0.28	1	1	1	1	1
W (μ m)	33.48	8.28	0.48	1.36	5.92	23.68	2.76	5.92	11.84
Area (μ m ²)	14.73x2	8.28x2	0.144x2	0.381x2	5.92x2	23.68	2.76	5.92	11.84
V_{GS} (V)	0.8202	0.8923	0.966	1.107	0.999	0.999	0.8923	0.999	0.999
V_{DSAT} (mV)	95.17	214.6	195	224.6	195	195	214.6	195	195

IP = Input Pair, CM = Current Mirror, Casc = Cascode

$$CMIR_{min} = -V_{th,in} + V_{DS,cs} = -0.7816 + 0.25 = -0.1566 \text{ V}$$

$$CMIR_{max} = V_{DD} - V_{GS,in} - V_{DS,cs} = 2.5 - 0.8202 - 0.625 = 1.05 \text{ V}$$

$$V_{out,min} = V_{DS,ncascode} + V_{star,nCS} = 0.625 + 0.25 = 0.875 \text{ V}$$

$$V_{out,max} = V_{DD} - V_{DS,pcascode} - V_{star,pCS} = 2.5 - 0.625 - 0.25 = 1.625 \text{ V}$$

$$\text{Max Swing} = 2(1.625 - 0.875) = 1.5 \text{ Vpp}$$

Cascode Bias Voltage Generation

The Total bias currents should be $< 5 \mu\text{A}$

NMOS Cascode Bias (V_{CASCN})

The NMOS cascode bias voltage is generated as:

$$V_{CASCN} = V_{GS,M_3} + V_{DS,M_2}$$

$$V_{CASCN} = V_{GS,M_{B1}} + V_{DS,M_{B2}} = 0.966 + 0.5 = 1.466 \text{ V}$$

The magic battery consists of two stacked devices. The magic battery consists of a diode-connected NMOS transistor M_{dcn} matched to the NMOS cascode device (M_{3A}/M_{3B}) while M_{B1} 's $V_{GS} = V_{cascode,N}$ which is sized using a vgs sweep vs gm/id where the chosen current is $1 \mu\text{A}$.

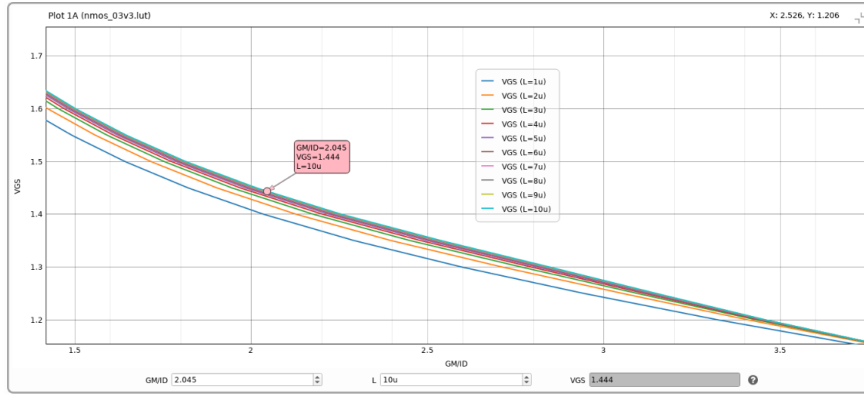


Figure 13 NMOS cascode bias generator

$L=10 \mu\text{m}$ & $W=280 \text{ nm}$

Table 4: V_{CASCN} bias generator

Parameter	M_{dc} (Diode)	M_{B1} (Stacked)	M_{B2} (PMOS CS)
Type	NMOS	NMOS	PMOS
Role	Matched to M_{3A}/M_{3B}	Provides V_{DS} drop	Current Source
I_D (μA)	1	1	1
L (μm)	0.35	10	1
W (μm)	0.48	0.28	0.592
Matching	M_{3A}/M_{3B}	—	M_9/M_8

PMOS Cascode Bias (V_{CASCP})

The PMOS cascode bias voltage is generated as:

$$V_{CASCP} = V_{DD} - |V_{GS,M4}| - V_{DS,M5}$$

$$V_{CASCP} = 2.5 - 1.107 - 0.625 = 0.768 \text{ V} \approx 0.77 \text{ V}$$

M_{dcp} is a diode-connected PMOS matching M4A/M4B, and M_{b2} is the PMOS generating $V_{cascode,p}$.

$$V_{cascode,p} = 0.77 \text{ V}$$

$$V_{DD} - V_{GS,Mb2} = V_{cascode,p}$$

$$V_{GS,Mb2} = 1.732 \text{ V}$$

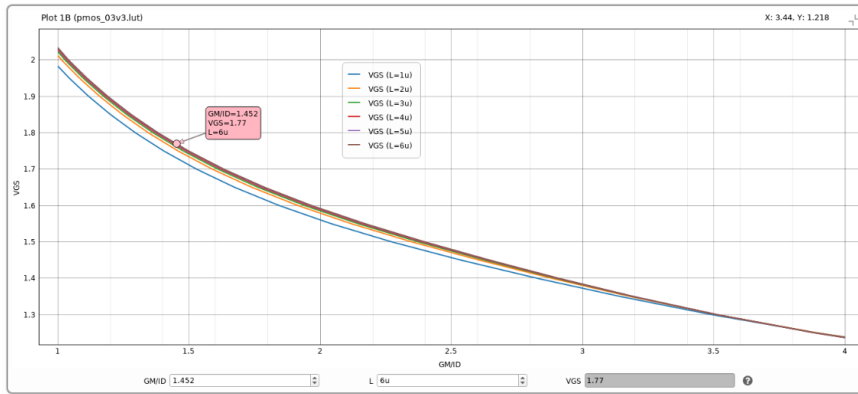


Figure 14 PMOS cascode bias generator

$L=6 \mu\text{m}$ & $W=560 \text{ nm}$

Table 5: 28 bias generator

Parameter	M_{dc} (Diode)	M_{b1} (Stacked)	M_{b2} (NMOS CS)
Type	PMOS	PMOS	NMOS
Role	Matched to M4A/M4B	Provides V_{GS} drop	Current Source
I_D (μA)	1	1	1
L (μm)	0.28	6	1
W (μm)	1.14	0.56	0.276
Matching	M4A/M4B	—	M7

Part 3: Open-Loop OTA Simulation

(Behavioral CMFB)

1) DCOP Analysis

We don't need high gain in the CMFB loop since its purpose is to set the common-mode voltage not to amplify signals so a gain of 1 in the error amplifier is sufficient.

$$V_{icm} = \frac{0 + 1}{2} = 0.5 \text{ V}$$

$$V_{ref} = \frac{V_{DD}}{2} = 1.25 \text{ V (For symmetric and max swing)}$$

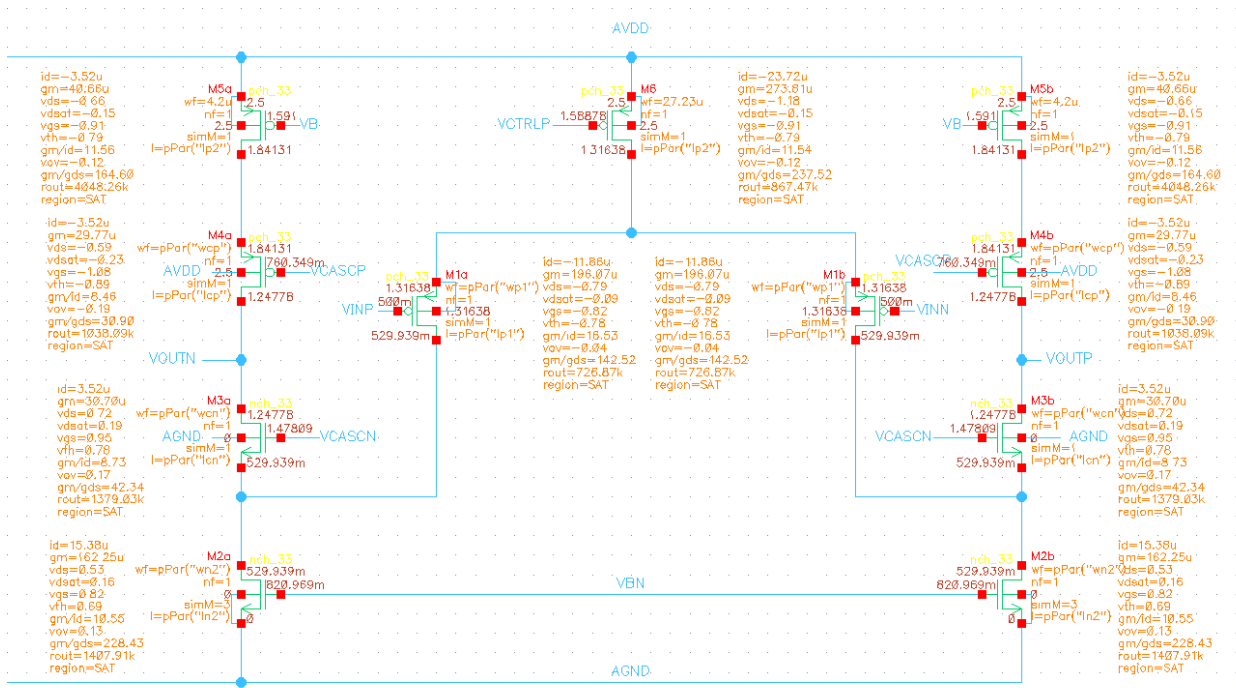
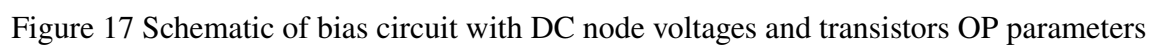


Figure 15 Schematic of the OTA with DC node voltages and transistors OP parameters



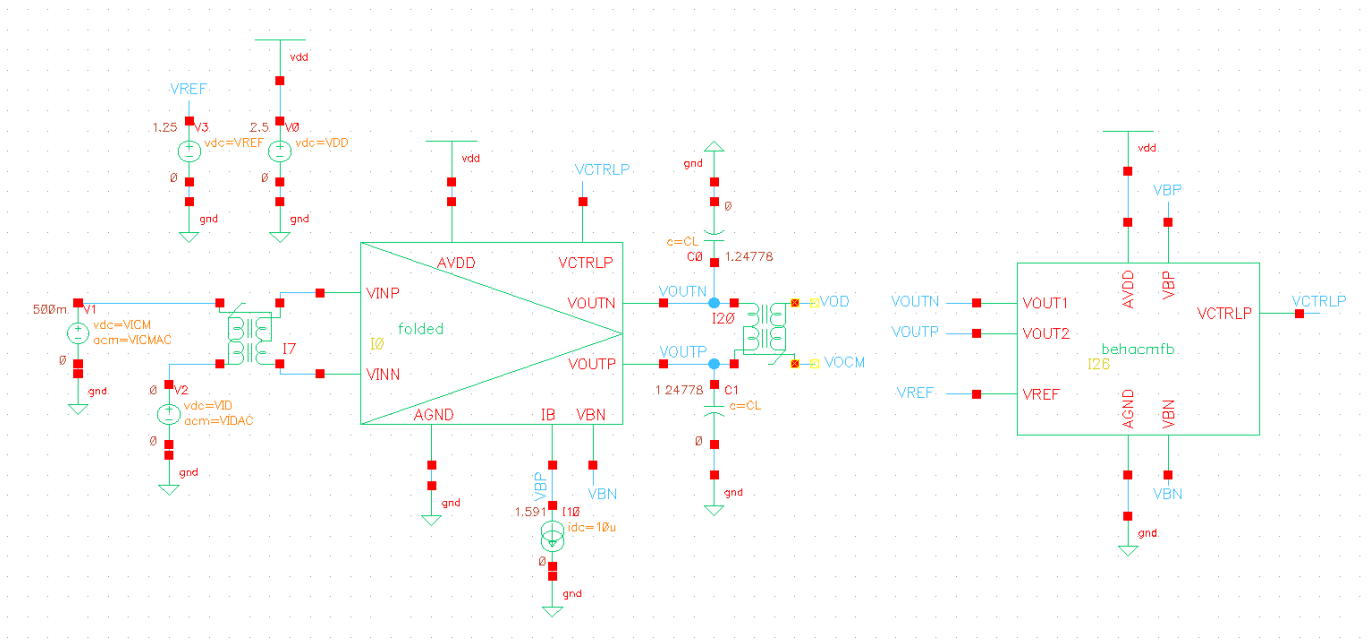


Figure 18 Folded cascode OTA with behavioral CMFB TB with DC Voltages annotated

CM Level at OTA Output

The CM level at the OTA output is set by V_{REF} :

$$V_{OCM} = V_{REF} = 1.25 \text{ V}$$

From Figure 32 $V_{ocm} = 1.247$

V_{REF} is selected to maximize the symmetrical output swing @ $V_{DD}/2$

The CMFB minimizes the error between common mode dc level of Vout and Vref.

Error Amplifier Inputs/Outputs

The differential input and output voltages of the error amplifier are:

$$V_{id,EA} = V_{CM} - V_{REF} = 1.24778 - 1.25 = -2.22 \text{ mV}$$

$$V_{od,EA} = A_{v,EA} \times V_{id,EA} = V_{od,EA} = -2.22 \text{ mV}$$

Since the CMFB loop gain is 1 the error amplifier output controls the PMOS current source.

2) Differential Small Signal

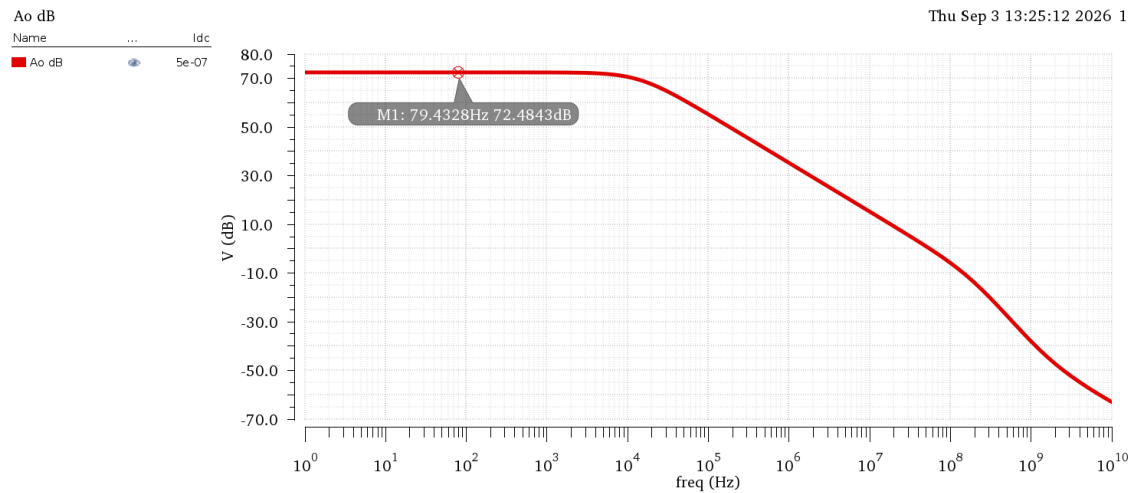


Figure 19 Differential gain Bode plot mag & phase (behavioral CMFB)

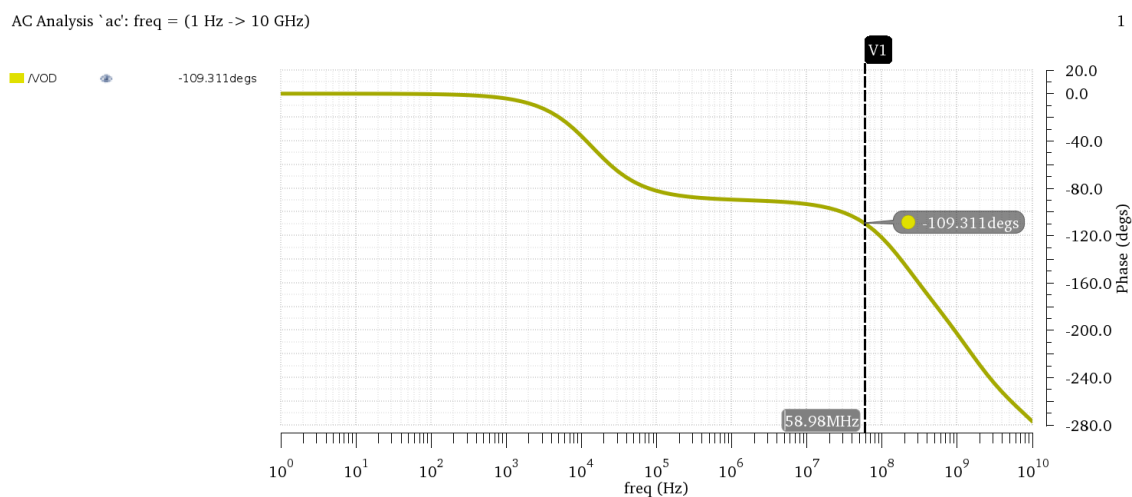


Figure 20 Differential gain Bode plot mag & phase (behavioral CMFB)

Ao_DC	4.209k	
BW	14.01k	
GBW	58.98M	
AodB DC	72.48	
UGF	57.25M	

Figure 21 AC simulation results (behavioral CMFB)

Hand Analysis

$$R_{out} = ((29.77+6.43)\mu \cdot 1.038M \cdot 4.048M) \parallel ((30.7+6.79)\mu \cdot 1.38M \cdot (1.41M \parallel 726.87k)) = 21.34 \text{ M}\Omega$$

$$A_{vd} = g_{m,1} \cdot R_{out} = 196.07 \times 21.34 = 4184.33 \text{ V/V}$$

$$A_{vd} = 72.43 \text{ dB}$$

$$BW_{OL} = \frac{1}{2\pi R_{out} C_L} = \frac{1}{2\pi \times 21.34 \times 10^6 \times 0.5 \times 10^{-12}} = 14.92 \text{ kHz}$$

$$GBW = BW_{OL} \times A_{vd} = 14.92 \times 10^3 \times 4184.33 = 62.43 \text{ MHz}$$

$$PM = 90^\circ - \tan^{-1} \left(\frac{BW_{OL}}{GBW} \right) \approx 89.98^\circ$$

Parameter	Hand Analysis	Simulation
DC Gain (dB)	72.43	72.48
GBW (MHz)	62.4	58.98
BW (MHz)	14.92	14.01
UGF (MHz)	62.4	57.25
Phase Margin (°)	70.67	89.98

Table 6: Differential gain comparison (behavioral CMFB)

Note: All results from Part 3 to 6 are based on the final optimized sizing, which is reported in Table 15.

Part 4: Open-Loop OTA Simulation

(Actual CMFB)

CMFB Circuit Design

CD Buffer Design:

The CD (source follower) buffers are used to avoid loading the OTA output with the sensing resistors. $I_B \geq I_x$

$$I_x = \frac{V_{od}}{2R_{sens}} = \frac{0.6}{100k} = 6 \mu A$$

$$I_B \geq 6 \mu A$$

For lower power consumption $I_B = 6 \mu A$

$$A_v = g_{m,CD} \cdot (R_{sensing} \parallel \frac{1}{g_{m,CD}})$$

Assumed $g_m/id = 15$ & $L=2 \mu m$

From ADT SA: $W=37.92 \mu m$, $V_{GS} = 845.9 mV$

$$V_{out,min} = V_{star,ncascode} + V_{DS,CS} = 0.75 V$$

Maximum output will be limited by CMFB

$$V_{out,max} = V_{DD} - V_{star,CS} - |V_{GS,CD}| = 2.5 - 0.2 - 0.8459 = 1.4541 V$$

$$V_{ref} = \frac{V_{out,max} + V_{out,min}}{2} = 1.1 V$$

Output swing is $2(1.4541-0.75) = 1.4 V_{pp}$

Error Amplifier Design

Input Pair (NMOS since output is closer to VDD):

$$V_{DS} = 0.8 \text{ V}, \quad I_D = 1 \text{ } \mu\text{A}, \quad gm/id = 15, \quad L = 1 \text{ } \mu\text{m}$$

From ADT SA: $W=1.44 \text{ } \mu\text{m}$, $V_{GS} = 742.2 \text{ mV}$, $V_{star} = 135.8 \text{ mV}$, $V_{th} = 687 \text{ mV}$

The width and length were reduced by a factor of 2.5 to improve the phase margin.

L and gm/ID for some transistors are already known, why?

The NMOS current source biasing the error amplifier and the PMOS current sources biasing the common-drain buffers were sized previously; only their widths are scaled using current ratios while L and gm/I_D remain unchanged. Additionally, the diode-connected loads in the error amplifier are matched to the PMOS current mirrors to minimize the error between V_{ctrlp} and V_{ocm} when the CMFB loop is open.

Design of Shifting Branch

Due to the $5 \text{ } \mu\text{A}$ current restriction for the three bias branches, $1 \text{ } \mu\text{A}$ is allocated to the CD + CS branch for $V_{ref} + V_{GS}$ (with V_{CASCN} and V_{CASCP} already using $1 \text{ } \mu\text{A}$ each). The current source is sized by scaling the existing PMOS current source by $1/10$. The common-drain transistor matches the previously sized CD (M4/M5) in V_{GS} and L .

$$I_D = 1 \text{ } \mu\text{A}, \quad V_{GS} = 845.9 \text{ mV}, \quad L = 1 \text{ } \mu\text{m}, \quad \text{From SA: } W = 6.2 \text{ } \mu\text{m}$$

Table 7: Shifting Branch

Parameter	Common Drain (CD) M9	Current Source (CS) M10
Type	PMOS	PMOS
I_D (μA)	1	1
L (μm)	1u	1
W (μm)	6.2	0.276
Matching	M4/M5	M9/M8

Current Budget Summary

Table 8: Bias branches current consumption

Branch	Current (μA)
V_{CASCN}	1
V_{CASCP}	1
Shifting Branch	1
Total	3

Total Bias branches Current = $3 \mu\text{A} < 5 \mu\text{A}$

Table 9: CMFB transistor sizing

Parameter	M4/M5	M6/M7	M8	M10	M2/M3	M0/M1	M9
Type	PMOS (CD_1)	NMOS (EA)	NMOS (CS)	PMOS (CD_2)	PMOS (DL)	PMOS (CS)	PMOS (CD_2)
Role	SF	EA IP	EA CS	CD CS	EA Load	CD_1 CS	CD CS
I_D (μA)	6	1	2	1	1	6	1
g_m/I_D (S/A)	15	15	10	10	10	10	13
L (μm)	1	0.4	1	1	1	0.5	1
W (μm)	37.92	0.576	0.552	1.1	1	3.552	6.2

CD = Common Drain, EA = Error Amplifier, CS = Current Source, DL = Diode Load, IP = Input Pair, SF = Source Follower

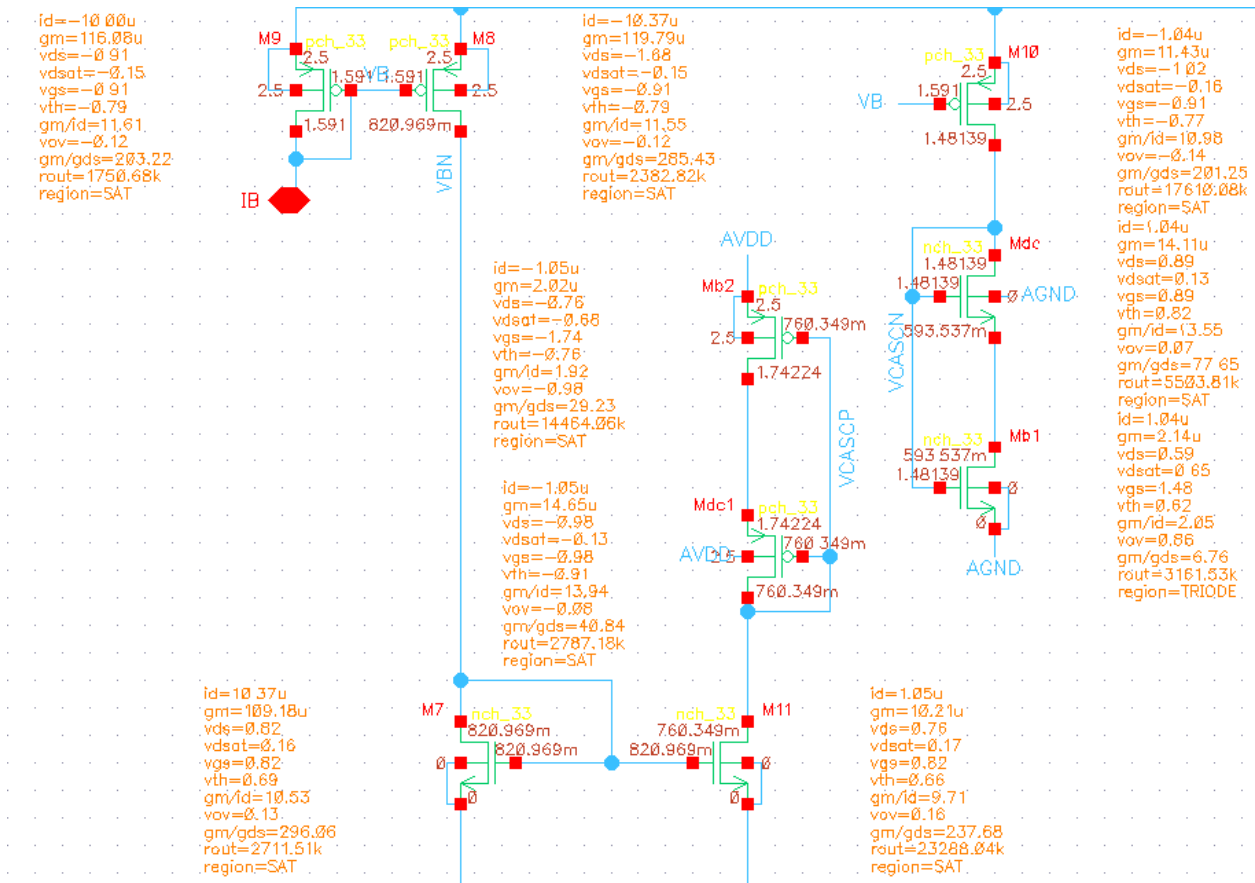


Figure 24 Schematic of bias circuit with DC node voltages and transistors OP parameters

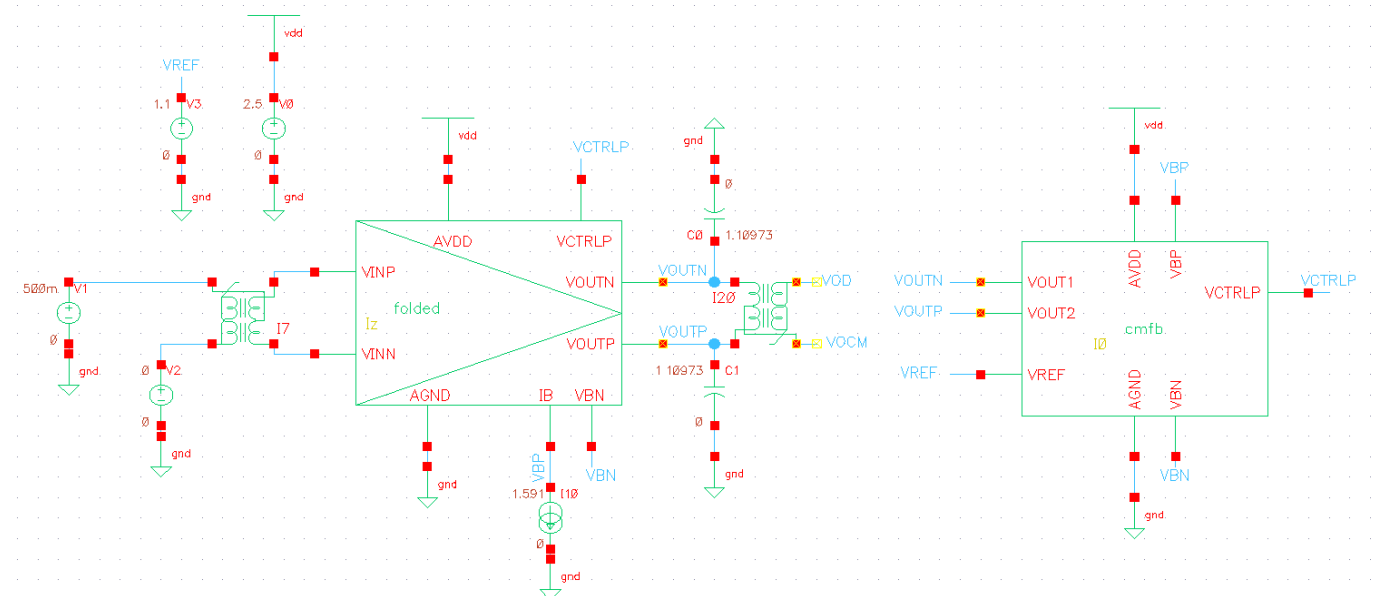


Figure 25 Folded cascode OTA with behavioral CMFB TB with DC Voltages annotated

CM Level at OTA Output

The CM level at the OTA output is set by the CMFB loop to:

$$V_{OCM} = V_{REF,new} = 1.109 \text{ V}$$

$V_{REF,new}$ is selected to maximize the symmetrical output swing based on the new output range:

$$V_{out,min} = 0.75 \text{ V}, \quad V_{out,max} = 1.454 \text{ V}$$

$$V_{REF,new} = \frac{V_{out,min} + V_{out,max}}{2} = \frac{0.75 + 1.454}{2} = 1.102 \text{ V} \approx 1.1 \text{ V}$$

Error Amplifier Inputs/Outputs

The differential input and output voltages of the error amplifier are:

$$V_{id,EA} = V_{CM} - V_{REF,new}$$

$$V_{od,EA} = A_{v,EA} \times V_{id,EA}$$

From simulation:

$$V_{id,EA} = 1.95477 - 1.94922 = 5.55 \text{ mV}$$

$$V_{od,EA} = 1.58143 - 1.58882 = -7.39 \text{ mV}$$

$$A_{v,EA} = \frac{g_{m,1}}{g_{m,2}} \approx 1.38$$

Differential Small Signal

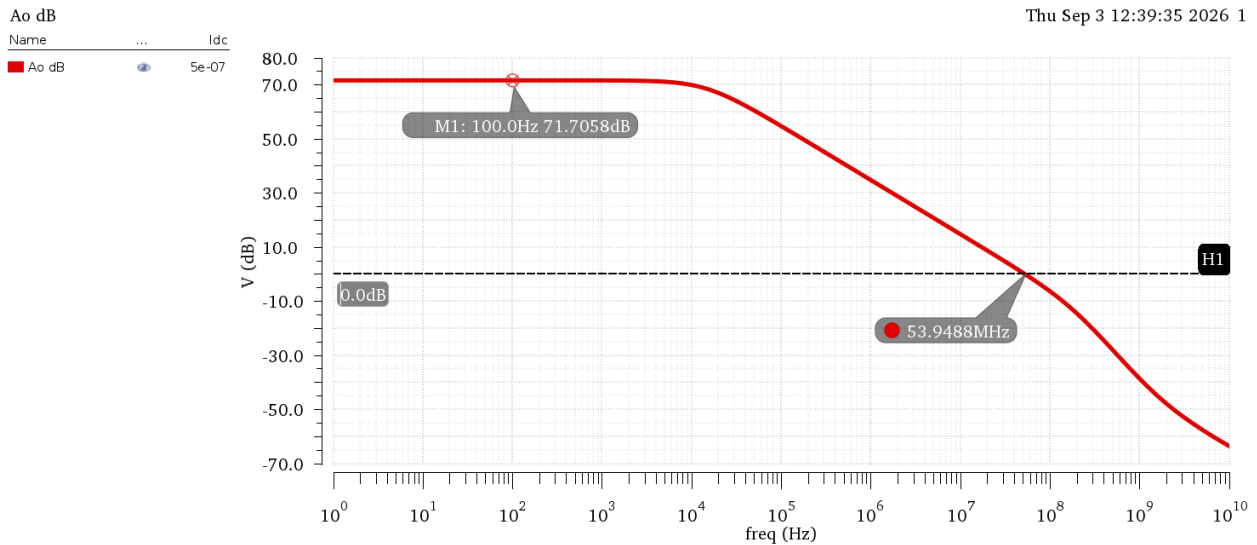


Figure 26 Differential gain Bode plot mag dB (actual CMFB)

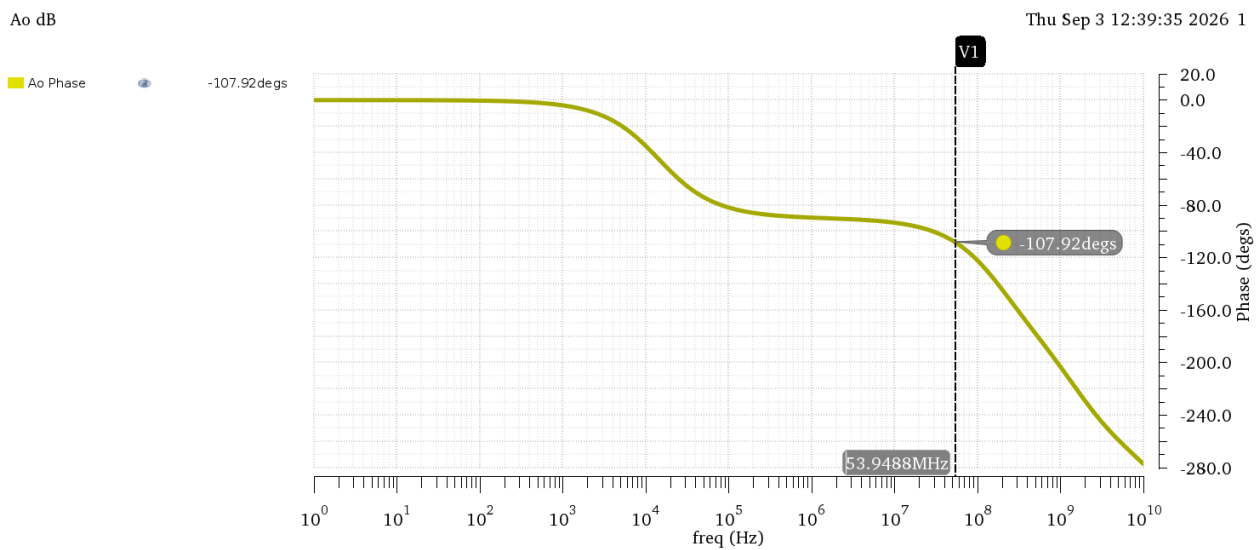


Figure 27 Differential gain Bode plot phase (actual CMFB)

Ao_DC	3.849k	
BW	14.51k	
GBW	55.83M	
AodB DC	71.71	
UGF	54.29M	

Figure 28 AC Simulation results

$$R_{out} = ((g_{m,4} + g_{mb,4}) \cdot r_{o4} \cdot r_{o5}) \parallel ((g_{m,3} + g_{mb,3}) \cdot r_{o3} \cdot (r_{o2} \parallel r_{o1}))$$

$$R_{out} = ((30.03 + 6.43)\mu \cdot 4.06M \cdot 1.1M) \parallel ((30.49 + 6.79)\mu \cdot 1.23M \cdot (1.4M \parallel 728k)) = 19.35 \text{ M}\Omega$$

$$A_{vd} = g_{m,1} \cdot R_{out} = 196.04 \times 19.35 = 3793.374 \text{ V/V}$$

$$A_{vd}[\text{dB}] = 20 \log_{10}(A_{vd}) = 71.58$$

$$BW_{OL} = \frac{1}{2\pi R_{out} C_L} = \frac{1}{2\pi \times 19.35 \times 10^6 \times 0.5 \times 10^{-12}} = 16.45 \text{ kHz}$$

$$GBW = BW_{OL} \times A_{vd} = 16.45 \times 10^3 \times 3793.374 = 62.4 \text{ MHz}$$

$$PM = 90^\circ - \tan^{-1} \left(\frac{BW}{GBW} \right) = 89.98^\circ$$

Table 10: Comparison between hand calculations and simulation results

Parameter	Hand Calculation	Simulation
A_{vd} (V/V)	3793.374	3849
A_{vd} (dB)	71.58	71.7
BW_{OL} (kHz)	16.45	14.51
GBW (MHz)	62.43	55.83
PM ($^\circ$)	89.98	72.08

Part 5: Closed Loop Simulation

OP Analysis

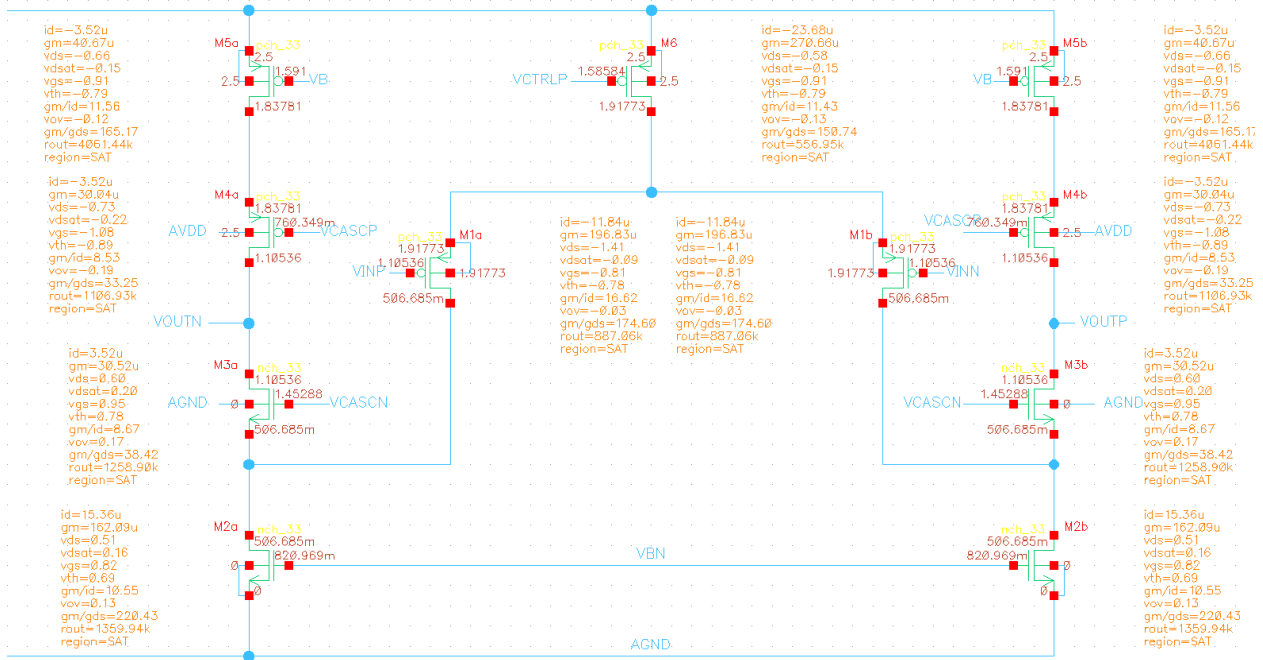


Figure 29 Schematic of the OTA with DC node voltages and transistors OP parameters

$$\text{Total Amplifier Current} = 30.72 \mu\text{m}^2$$

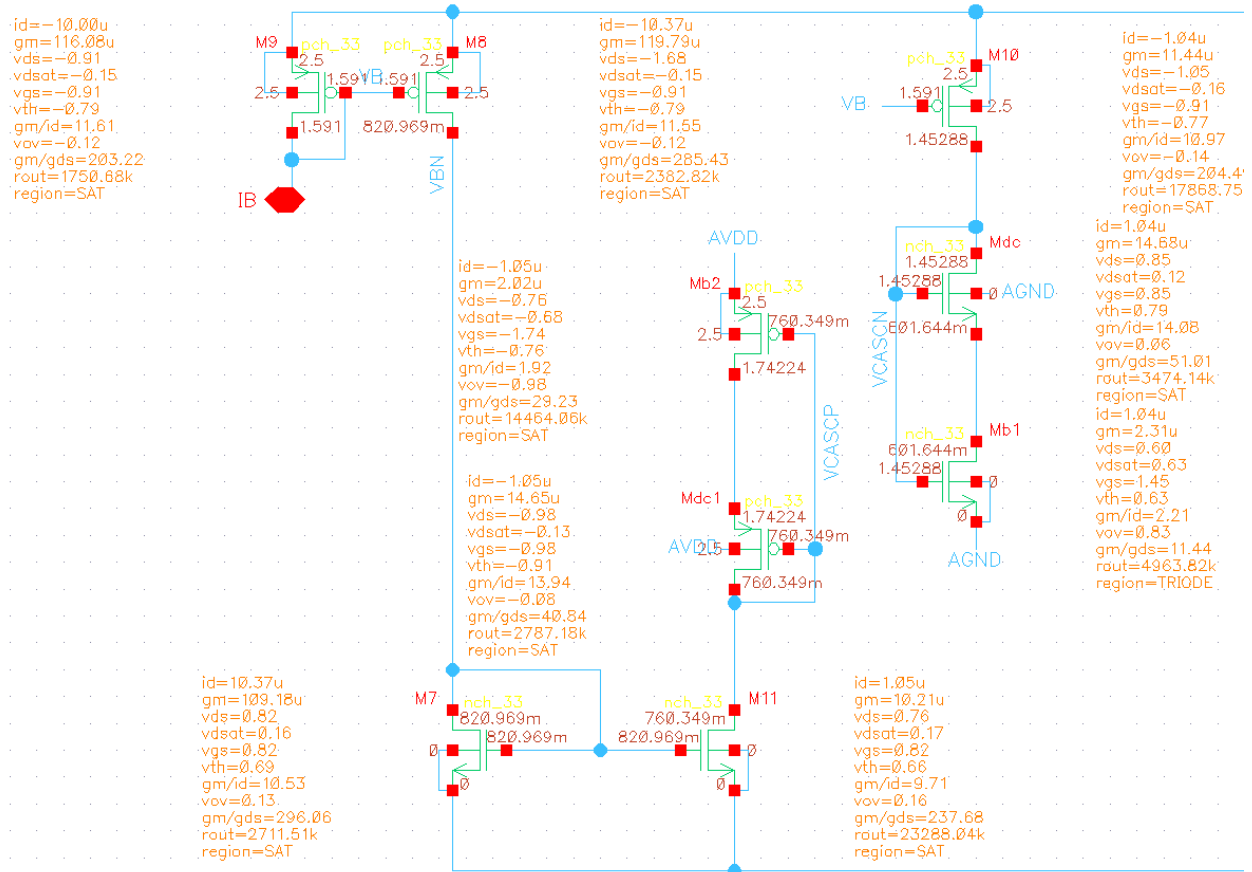


Figure 30 Schematic of bias circuit with DC node voltages and transistors OP parameters

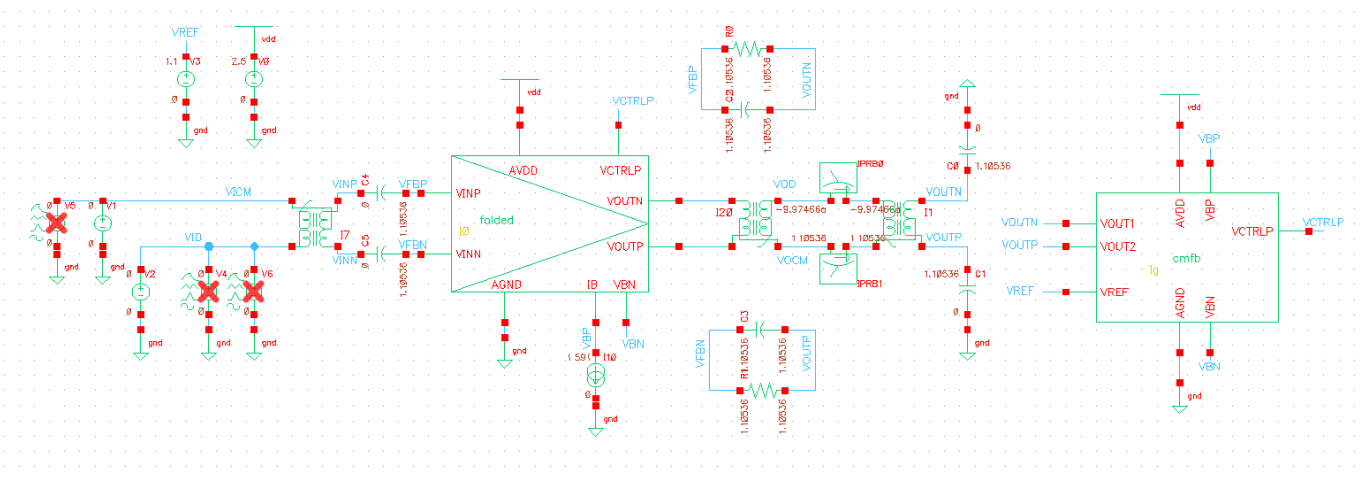


Figure 31 Schematic of Actual CMFB with DC Voltages annotated and transistors OP parameters

Differential Closed-Loop Response

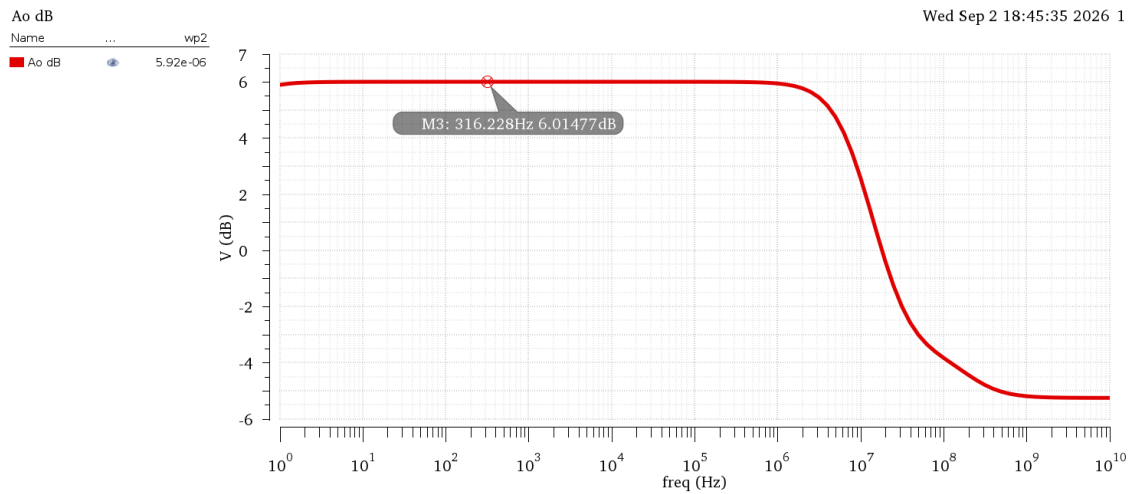


Figure 33 Closed-loop differential gain mag in dB vs frequency

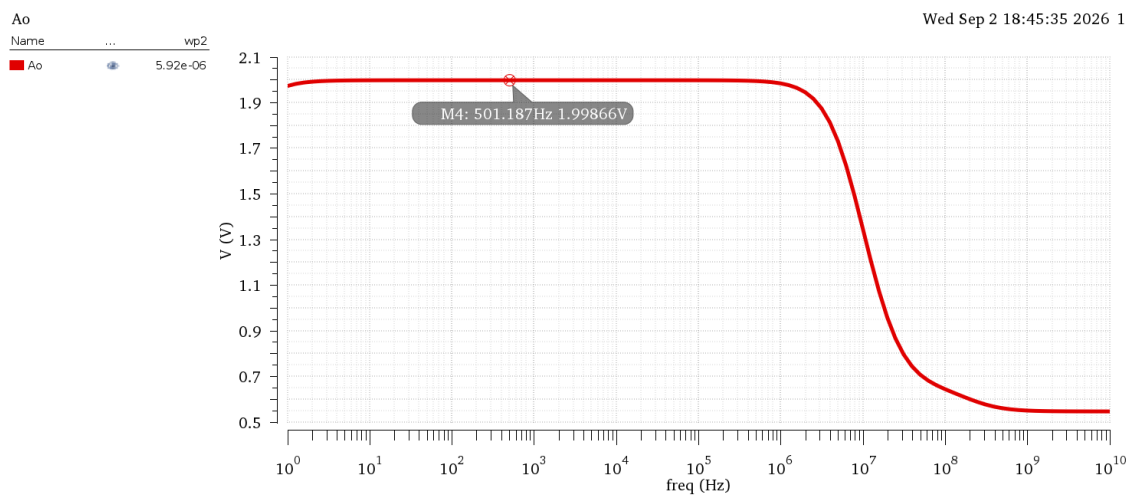


Figure 34 Vodiff mag vs frequency

Ao @DC	1.998
Ao dB @DC	6.014
CL GBW	19.32M
CL BW	9.765M

Figure 35 AC Simulation results

Table 11 Closed-loop AC results

Parameter	Value	Unit
DC Gain	6.014	dB
CL BW	9.765	MHz
CL GBW	19.32	MHz

STB Analysis

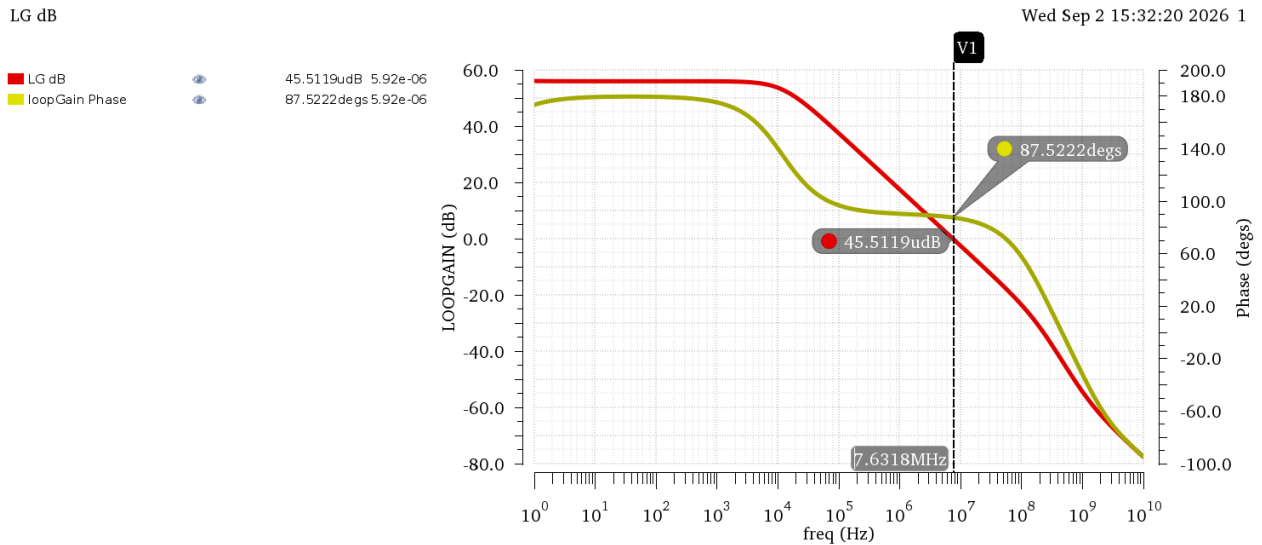


Figure 36 Loop gain magnitude and phase for diff loop

Ao @DC	1.998	
Ao dB @DC	6.014	
CL GBW	19.32M	
CL BW	9.765M	
LG dB @DC	61.96	
LG @DC	1.254k	
PM	87.5	

Figure 37 STB Simulation results for diff loop

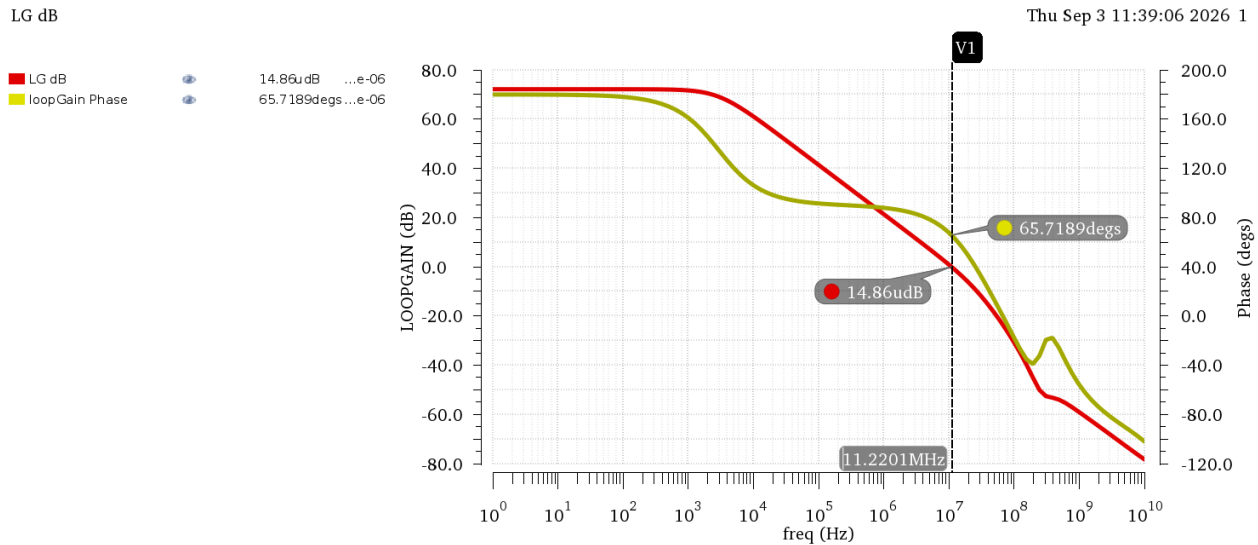


Figure 38 Loop gain magnitude and phase for CM loop

Ao @DC	1.998	
Ao dB @DC	6.014	
CL GBW	19.32M	
CL BW	9.765M	
LG dB @DC	72.43	
LG @DC	4.184k	
PM	65.66	

Figure 39 STB Simulation results for CM loop

Comment:

Both differential and CM loops are stable with adequate phase margin. The differential loop gives a higher phase margin of 87.5° , ensuring an over-damped transient response. The CM loop has a lower phase margin of 65.66° due to the limited bandwidth of the CMFB circuit, which introduces additional poles. This is still acceptable as it provides sufficient stability margin.

The DC loop gain of the differential loop (61.96 dB) is lower than the CM loop (72.43 dB) because the CMFB error amplifier provides additional gain in the CM path. The GBW of both loops is identical (19.32 MHz) since it is primarily determined by the OTA's dominant pole.

Comment: The DC loop gain from STB (61.96 dB) is lower than the open-loop gain (71.71 dB) due to the loading effect of the feedback network. The GBW is reduced from 55.83 MHz (open-loop) to 19.32 MHz (STB), consistent with the closed-loop bandwidth being limited by the feedback network and the increased effective load capacitance.

$$LG = A_{OL}(dB) + 20\text{Log}(\beta) = 71.71 - 9.542 = 62.168\text{dB}$$

$$C_{L,eff} = C_L + C_F \parallel C_{IN}$$

$$GBW_{CL} = GBW_{OL} \times \frac{C_{IN}}{C_{IN} + C_F}$$

$$GBW_{CL} = 55.83 \times \frac{2}{3} = 37.22 \text{ MHz}$$

The simulated STB GBW is 19.32 MHz, which is lower than this theoretical value. Since the feedback network loads the output with C_F and C_{in} in series together parallel to the load increasing the total load capacitance to approximately:

$$C_{L,total} = C_L + \frac{C_{IN} \cdot C_F}{C_{IN} + C_F} = 0.5 + \frac{2}{3} = \frac{7}{6} \text{ pF}$$

$$GBW_{CL} = \frac{g_m}{2\pi C_{L,total}} \times \frac{C_F}{C_{IN} + C_F}$$

$$GBW_{CL} = \frac{196 \times 10^{-6}}{2\pi \times \frac{7}{6} \times 10^{-12}} \times \frac{2}{3} \approx 18 \text{ MHz}$$

Table 12: Comparison of differential loop parameters: STB vs. open-loop simulation

Parameter	STB (Diff Loop)	Open-Loop	Calculated
DC LG (dB)	61.96	71.71	62.17
GBW (MHz)	19.32	55.83	17.81

Part 6: Transient Analysis

Differential and CMFB loops stability

1) Differential Input Pulse

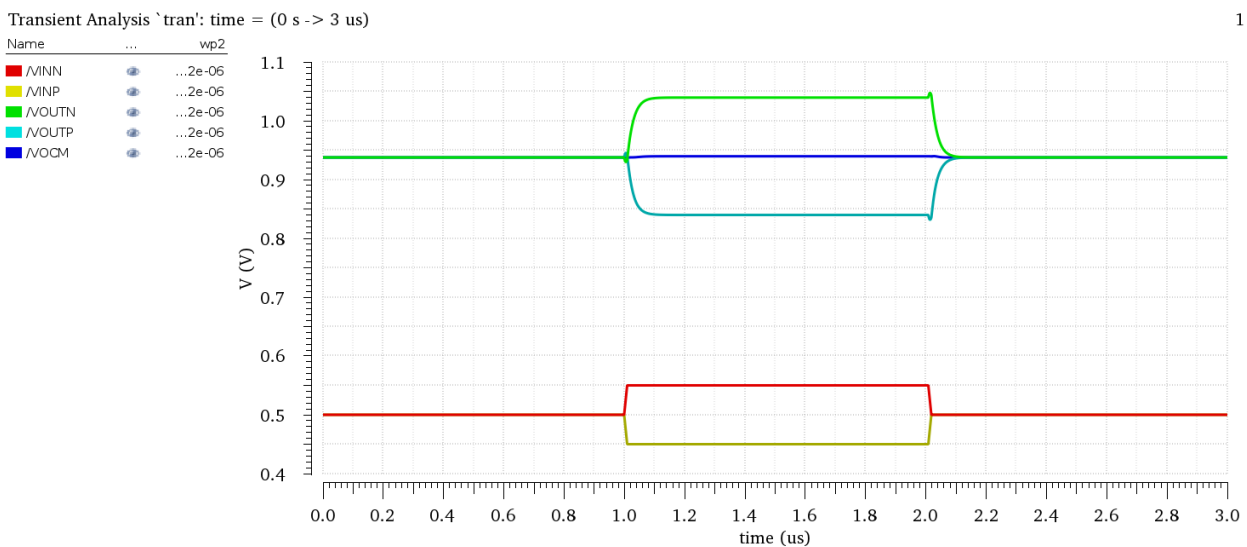


Figure 40 Transient response to differential input pulse



Figure 41 settling time simulation value

Possible Design Changes (if spec not met):

- Increase the bias current of the input pair to improve the slew rate.
- Increase the GBW by increasing $g_{m,1}$ (larger input pair W).
- Reduce the feedback factor or adjust C_F/C_{IN} ratio.
- Improve the CMFB loop bandwidth by increasing the error amplifier current.

After increasing g_m by increasing width of the input stage to $35\ \mu\text{m}$, and increasing the split ratio as previously shown in OP to 3.36 from 2 and decreasing the PMOS current mirror's area since they have high parasitic caps to adjust settling time.

Settling T	98.67n
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Figure 42 settling time simulation value after optimization

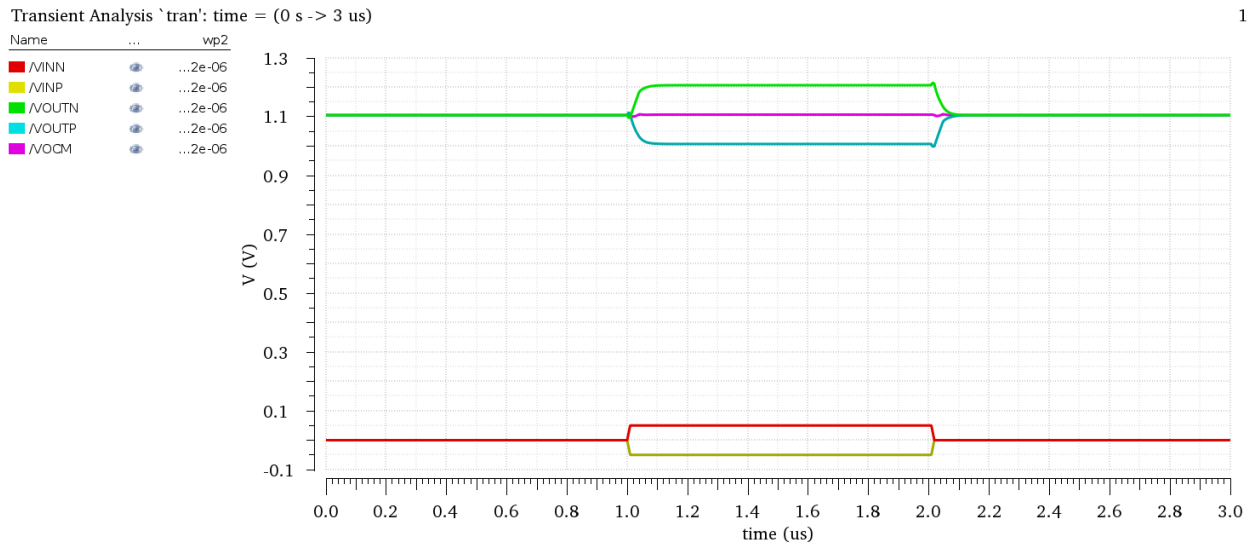


Figure 43 Transient response to differential input pulse

Observations:

1. The glitches observed at the output are caused by the feedforward path of the feedback capacitors (C_F), which allow high-frequency signal transitions to leak directly to the output. These glitches can be reduced by increasing the rise and fall times of the input step.
2. No differential ringing occurs as $PM_{diff} = 87.5^\circ > 75^\circ$.
3. Slight CM ringing is present since $PM_{CM} < 75^\circ$ but settles quickly due to high GBW.
4. Impulses at the start and end of the CM transient arise from initial conditions, capacitive coupling and slewing of the CMFB error amplifier.

2) CM Input Pulse

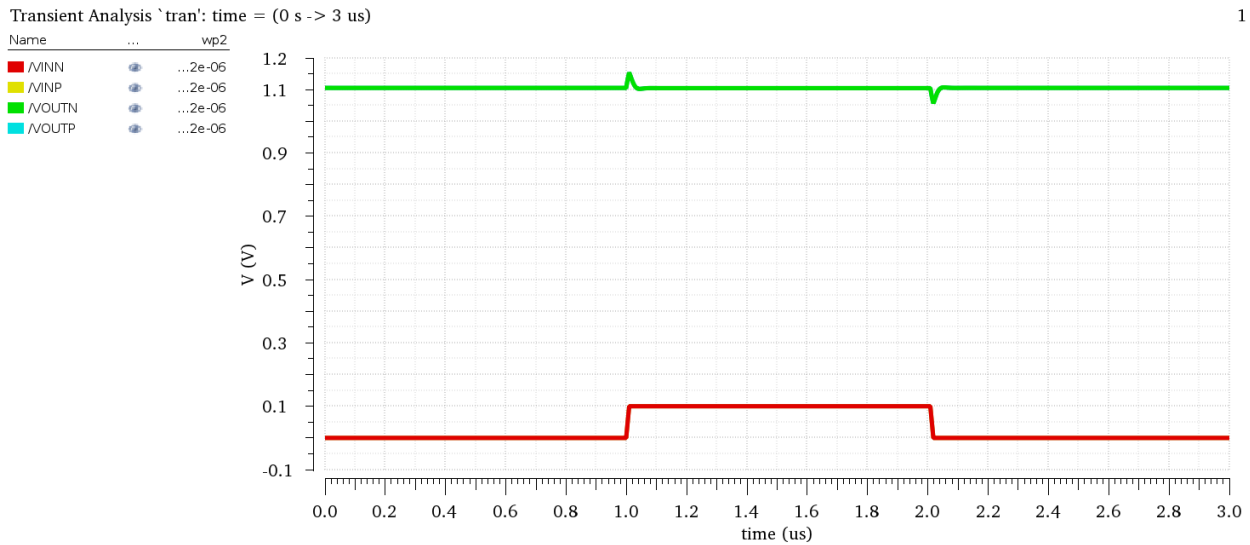


Figure 44 Transient response to CM input pulse

Observations:

1. The differential output remains balanced with no significant differential ringing which indicates good differential loop stability.
2. The CM loop gives slight ringing at the transitions due to the limited phase margin (65.66°) but it settles within the required time. Both loops are stable with adequate phase margin.

3) Output Swing

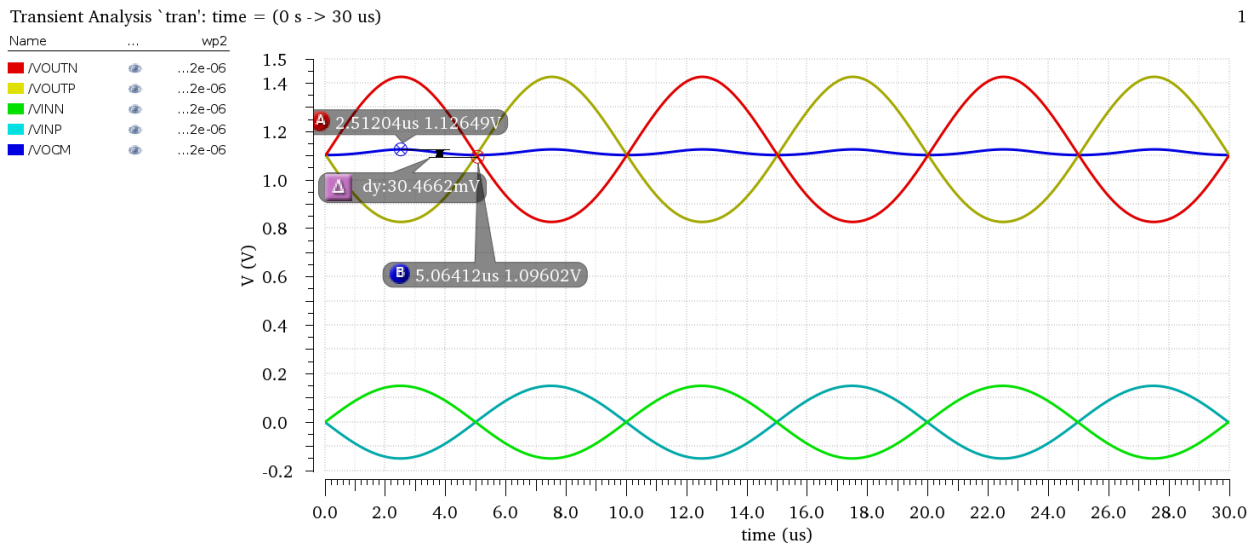


Figure 45 VINP, VINN, VOUTP, VOUTN, and VOCM overlaid with differential sinusoidal input

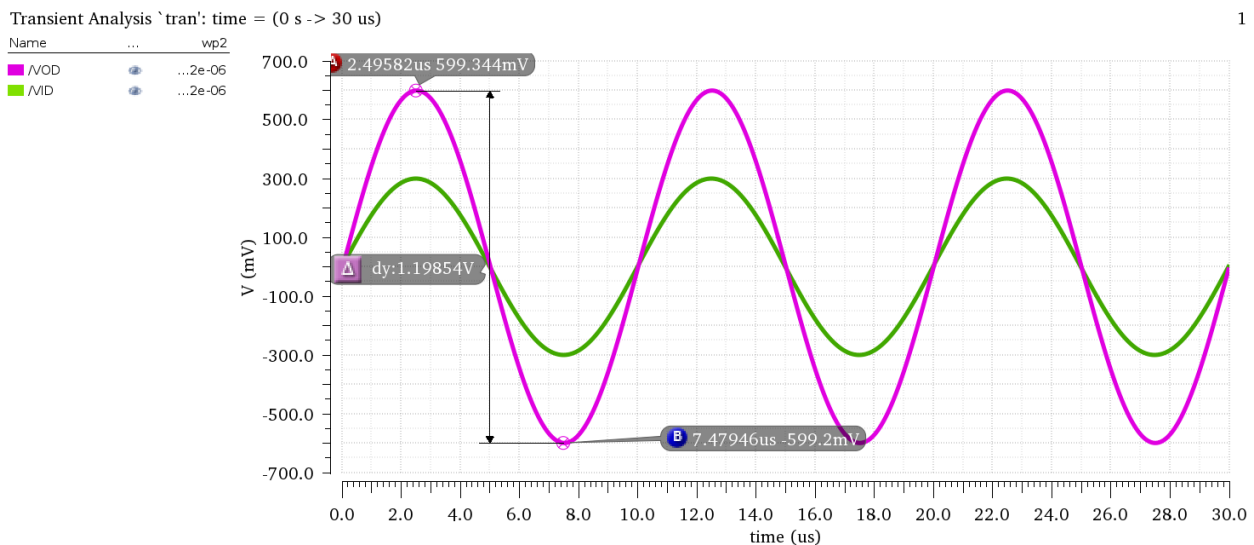


Figure 46 V_{IDIFF} and V_{ODIFF} overlaid vs time

Parameter	Value	Unit
Input Peak-to-Peak Swing	600	mV
Output Peak-to-Peak Swing	1.199	V
Closed Loop Gain	1.996	V/V

Table 13: Output swing results

Conclusion

Parameter	Specification	Result
Closed Loop Gain	2	1.998 V/V
Phase Margin Diff	$\geq 70^\circ$	87.5°
Phase Margin CM	$\geq 70^\circ$	70°
CMIR Low	≤ 0 V	-0.15 V
CMIR High	≥ 1 V	1.06 V
Differential Output Swing	1.2 V _{pk-to-pk}	1.4 V _{pk-to-pk}
DC Loop Gain	≥ 60 dB	61.96 dB
Settling Time	≤ 100 ns	98.63 ns

Table 14: Design summary

Final Sizing Table

Table 15: Folded cascode OTA transistor parameters

Parameter	M1A/M1B	M2A/M2B	M3A/M3B	M4A/M4B	M5A/M5B	M6	M7	M8	M9
Type	PMOS	NMOS	NMOS	PMOS	PMOS	PMOS	NMOS	PMOS	PMOS
Role	IP	NMOS CM	Casc N	Casc P	PMOS CM	PMOS CM	NMOS CM	PMOS CM	I_{ref}
I_D (μ A)	10	15	5	5	5	20	10	5	10
L (μ m)	0.4	1	0.35	0.28	0.5	0.5	1	0.5	0.5
W (μ m)	35	4.14	0.48	1.14	1.48	23.68	2.76	1.48	5.92
Area (μ m ²)	14x2	4.14x2	0.168x2	0.32x2	0.74x2	11.84	2.76	0.74	2.96

IP = Input Pair, CM = Current Mirror, Casc = Cascode

$$\text{Total Amplifier Current} = 30.72 \mu\text{m}^2$$

$$\text{Total Area} = 57 \mu\text{m}^2$$

$$\text{FOM} = \frac{GBW}{I_{total} \times A_{total}} = \frac{58.98 \times 10^6}{30.72 \times 10^{-6} \times 57 \times 10^{-12}} = 3.368 \times 10^{22} \text{ Hz}/(\text{A} \cdot \text{m}^2)$$